# Serat M. Saad

Ph.D. Student in Astronomy, The Ohio State University

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## Research Interests

**Dynamics:** binary stars, radial migration, tests of Modified Newtonian Dynamics (MOND), an alternative to dark matter

**Stellar Spectroscopy:** chemical abundances, metal-rich stars, star spots, young stellar objects

**AI for Astronomy:** LLM agents, agent coordination, automation of astrophysical research

## Education

Ph.D. in Astrophysics (in progress), The Ohio State University 2025-Present
*Advisors: Prof. Yuan-Sen Ting and Prof. Krzysztof (Kris) Stanek*

A.B. in Physics & Applied Mathematics, Vanderbilt University 2021-2025
*Thesis: Understanding Accretion Activity in Young Stars using SDSS-V BOSS Spectra*
*Advisor: Prof. Keivan Stassun*

## Honors & Awards

University Distinguished Fellowship, The Ohio State University 2029-2030
*graduate-school fellowship supporting doctoral study*

University Fellowship, The Ohio State University 2025-2027
*graduate-school fellowship supporting doctoral study*

Hult Prize, National Finalist 2026
*for a student venture expanding access to personalized education*

Larry R. Cathey Award, Vanderbilt University 2025
*annual departmental award for the top graduating student studying astronomy*

Highest Honors in Astronomy, Vanderbilt University 2025
*for distinction in the undergraduate honors thesis and its defense*

International Full-ride Student Grant, Vanderbilt University 2021-2025
*full four-year scholarship covering all costs of undergraduate study*

Dean's List, Vanderbilt University 2021-2025
*eight semesters of academic distinction*

Buchanan Fellowship, Vanderbilt University 2022
*supported a data-science project applying NLP to British periodical collections*

Silver Medal, International Olympiad on Astronomy and Astrophysics 2020
*represented Bangladesh at IOAA 2019 and the Global e-Competition (GeCAA) 2020*

## Publication Summary

13 papers, 7 as first author, 8 published or accepted in refereed journals.
122 citations from 117 citing papers (SciX Library).

## Telescope Time Allocations

**Principal Investigator**

Large Binocular Telescope, PEPSI, 24 hours · PANTERA: ultra metal-rich stars (submitted) 2027A
Large Binocular Telescope, PEPSI, 20 hours · PANTERA: ultra metal-rich stars 2026B
Large Binocular Telescope, PEPSI, 20 hours · PASTA: 36 planet-hosting stars 2026A
Large Binocular Telescope, PEPSI, 28 hours · PASTA: 43 planet-hosting stars 2025B

**Co-Investigator**

Large Binocular Telescope, iLocater+PEPSI, 15 hours · close-in binaries (PI: M. Tala Pinto, submitted) 2027A
WIYN, NEID, NN-EXPLORE Key Program · 65 HWO target stars (PI: M. C. Johnson, submitted) 2026
Automated Planet Finder, Levy, 40 hours · metal-rich stars (PI: D. M. Rowan) 2026B
Chandra X-ray Observatory, Cycle 26 · Pink Mergers (PI: Prof. Adi Foord) 2024

## Invited Talks and Conference Presentations

Monday Science Seminar, University of Wisconsin–Madison 2026
*Title to be announced*

LLM Agents Seminar, NASA AI/ML STIG 2026
*Hands-on session II: agentic coding and research tools · recording · slides*

AAS 245 Winter Meeting, American Astronomical Society 2025
*An Alternative to the Supermassive Black Hole Binary Hypothesis for J1430 (Tick Tock)*

Cool Stars 22, Cambridge 2024
*Using Low Resolution Optical Spectra to Identify Young Stars and Their Properties*

IEEE Quantum Computing and Engineering (QCE) 2023
*Developing Workforce in Quantum Industry: The Wond'ry Quantum Studio*

## Teaching & Mentoring

*Teaching and grading in astronomy, statistics, data science, and programming, alongside mentoring olympiad students in Bangladesh and graduate-school applicants from underrepresented backgrounds.*

Mentor, Project SHORT 2025
*free admissions mentoring for underrepresented applicants to graduate and medical programs*

Mentor, Bangladesh Olympiad on Astronomy and Astrophysics 2021-2024
*national competition that selects and trains Bangladesh's team for the international olympiad (IOAA)*

TA, DS 2100: Statistics for Data Science, Vanderbilt University Spring 2025
TA, ASTR 1010: Introductory Astronomy, Vanderbilt University Spring 2024
Grader, DS 1100: Introduction to Data Science, Vanderbilt University Fall 2023
Instructor, Quantum Computing for Everyone, The Wond'ry Vanderbilt Fall 2023
Grader, CS 1100: Introduction to Problem Solving with Python, Vanderbilt University 2022-2023
Tutor, Vanderbilt Student Athletics 2022

## Other Experiences

*Building an educational AI initiative, communicating new research through writing, and several years training Bangladesh's astronomy olympiad team.*

Co-founder, PorAI 2026-Present
*educational AI initiative bringing personalized, tutor-like instruction to students and schools*

Author and Editor, Astrobites 2026-Present
Journal Referee (A&A) 2026-
Observer, Large Binocular Telescope, 5 nights 2026
Academic Team Member, Bangladesh Olympiad on Astronomy and Astrophysics 2021-2024
Co-lead, The Wond'ry Quantum Studio 2022-2023
*student-led quantum computing program at Vanderbilt's innovation center*

Team Leader, International Olympiad on Astronomy and Astrophysics 2022
*led Bangladesh's national team and served as a question moderator at the international round*

## First Author Publications and Preprints

7. **Serat M. Saad** & Yuan-Sen Ting (2026)
"More than Half of Recent Astronomy Papers are Written with Language-Model Assistance"
*arXiv:2609.10664*
6. **Serat M. Saad** & Yuan-Sen Ting (2026)
"Spectroscopic Binary Detection as Agent-Callable Tools: Detecting 40,000+ Main-Sequence Binary Candidates from SDSS DR19 APOGEE Spectra"
*The Open Journal of Astrophysics, Vol. 9*
5. **Serat M. Saad**, D. M. Rowan, K. Z. Stanek, et al. (2026)
"PANTERA. I. Hunting for the Most Metal-Rich Stars in the Solar Neighborhood with High-Resolution Spectroscopy"
*The Open Journal of Astrophysics, submitted*
4. **Serat M. Saad** & Yuan-Sen Ting (2026)
"No Gravitational Anomaly in Wide Binaries from Forward Modeling of 3D Orbits"
*The Open Journal of Astrophysics, Vol. 9*

3. **Serat M. Saad** & Yuan-Sen Ting (2026)
"High-Precision Differential Radial Velocities of C3PO Wide Binaries: A Test of Modified Newtonian Dynamics (MOND)"
*The Open Journal of Astrophysics, Vol. 9*
2. **Serat M. Saad**, Marina Kounkel, Keivan G. Stassun, et al. (2025)
"ABYSS III: Observing Accretion Activity in Young Stars through Empirical Veiling Measurements"
*The Astronomical Journal, Vol. 170, 87*
1. **Serat M. Saad**, Kaitlyn Lane, Marina Kounkel, Keivan G. Stassun, et al. (2024)
"ABYSS II: Identification of Young Stars in Optical SDSS Spectra and Their Properties"
*The Astronomical Journal, Vol. 167, 125*

## Co-Author Publications and Preprints

6. Yuan-Sen Ting & **Serat M. Saad** (2026)
"Armillary: A Label-Free Coordinate System for Stellar Spectra"
*The Open Journal of Astrophysics, submitted*
5. Qinghui Sun, **Serat M. Saad**, Yuan-Sen Ting, et al. (2026)
"Planets Around Solar Twins/Analogs (PASTA) III: Chemical Clock Relations in Planet-Hosting Solar Analogs"
*Astronomy & Astrophysics, accepted*
4. SDSS Collaboration, including **Serat M. Saad** (2026)
"The Twentieth Data Release of the Sloan Digital Sky Surveys"
*The Astrophysical Journal Supplement Series, submitted*
3. Zimo Cheng, Sharon X. Wang, Fan Liu, ... **Serat M. Saad**, et al. (2026)
"A Bayesian Search for Planet Engulfment Signatures in Solar Analogs"
*AAS Journals, submitted*
2. SDSS Collaboration, including **Serat M. Saad** (2026)
"The Nineteenth Data Release of the Sloan Digital Sky Survey"
*The Astrophysical Journal Supplement Series, Vol. 285, 9*
1. Yuan-Sen Ting, **Serat M. Saad**, Fan Liu, Yuting Shen (2026)
"Egent: An Autonomous Agent for Equivalent Width Measurement"
*The Open Journal of Astrophysics, Vol. 9*

## Non-Refereed Publications

6. **Serat M. Saad** (2026)
"'Total Eclipse of the Heart' in Astronomy"
*Astrobites, August*
5. **Serat M. Saad** (2026)
"Where Have All the Star Couples Gone?"
*Astrobites, August*
4. **Serat M. Saad** (2026)
"Cosmic Cannibalism: When Stars Eat Their Planets"
*Astrobites, May*
3. **Serat M. Saad** (2026)
"Wormholes Might Be More Real Than We Thought"
*Astrobites, April*
2. **Serat M. Saad** (2026)
"A Wandering Supermassive Black Hole Eating a Star"
*Astrobites, March*
1. **Serat M. Saad**
"Pixels and Persuasion"
*As Seen on TV, digital exhibition, Vanderbilt University Libraries*

## References


Krzysztof (Kris) Stanek, The Ohio State University — Ph.D. Advisor, stanek.32@osu.edu
*University Distinguished Scholar; Guggenheim Fellow (2022)*

Yuan-Sen Ting, The Ohio State University — Ph.D. Advisor, ting.74@osu.edu
*Inaugural Chair, NASA AI/ML Science and Technology Interest Group; Kavli Scholar (2026)*

Keivan G. Stassun, Vanderbilt University — Undergraduate Advisor, keivan.stassun@vanderbilt.edu
*National Medal of Science (2024); MacArthur Fellow (2024)*